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Are we in right path for mediation analysis? Reviewing the literature and proposing robust guidelines

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ABSTRACT

This research proposes guidelines for hypothesizing and assessing mediation concepts for tourism and hospitality research. The mediation analysis plays a significant role in theory development and advancement in social science disciplines and is significantly increasing in recent years. However, following questionable hypothesis development, statistical assessment, and interpretation of results cause biased and unreliable understanding of mediation analysis and questionable findings for theory building and development. Drawing from a systematic review of 536 mediation papers published in five recent years (2016–2020) in top tier tourism and hospitality journals, the findings revealed several critical issues in different mediation analysis steps, including hypothesis development, mediation assessment and interpretation of results. The results highlighted the common methodological mistakes and misconceptions for mediation analysis. This paper provides robust guidelines for mediation analysis by highlighting those methodological issues and has significant theoretical and methodological contributions in the tourism and hospitality field.

1. Introduction

The mediation model has become increasingly ubiquitous (Bullock et al., 2010) and plays a significant role in theory development and evolution of knowledge in the social sciences (Pieters, 2017; Wood et al., 2007). According to Preacher and Hayes (2008), a mediation model describes how, or by what means, an independent variable (*X*) affects a dependent variable (*Y*) through one or more potential intervening variables or mediators (*M*). The mediation effect has attracted scholars' attention and interest from various disciplines such as business, management, psychology, and education (Memon et al., 2018). The story is the same in tourism and hospitality, and studies referring mediators have increased dramatically in recent years, as depicted in Fig. 1. Despite the growing interest among scholars in mediation studies, some of the outdated approaches, such as the causal steps approach (Baron & Kenny, 1986) and the Sobel test (Sobel, 1982), are still applied by many to assess the mediation effect (Aguinis et al., 2017; Hayes, 2018). Moreover, due to unfamiliarity with mediation assessment, some critical issues about mediation hypothesis development, mediation assessment methods, inference approaches of indirect effect, and interpretation of mediation results have been continuously confounding some tourism and hospitality scholars.

Given the increasing popularity of mediation assessment and the above-mentioned critical issues, a systematic review that evaluates the status quo and thus guides future mediation studies is long overdue in tourism and hospitality. Therefore, by evaluating the current application of mediation analysis and clarifying the misunderstandings, this review is timely in advancing tourism and hospitality scholars' understanding of mediation assessment. Based on state-of-the-art literature, this review also recommends guidelines on articulating mediation hypotheses, which methods should be used for mediation analysis and indirect effect inference, and how to report and interpret mediation results.

2. Critical issues in mediation analysis

2.1. Mediation hypothesis development

Mediation effects are usually not explicitly hypothesized in many

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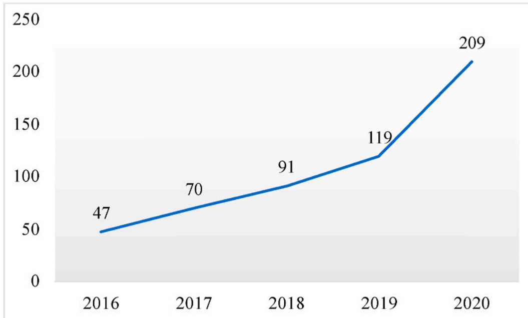

Fig. 1. Number of tourism and hospitality studies relating to mediation models.

studies (Hair et al., 2012). In Rungtusanatham et al. (2014)'s review of mediation studies in supply chain management, they noted that nearly 75% of the studies even did not develop mediation hypotheses. It should be emphasized that the process of mediation is a causal explanation in essence, which assumes that the mediator is causally located between an independent variable (*X*) and a dependent variable (*Y*) (Hayes, 2018). As theory is always the basis of empirical studies (Fisher & Aguinis, 2017; Hair et al., 2017), the theoretical foundation underlying the mediating relationships should be considered deliberately (MacKinnon et al., 2012). Merely arguing that the mediator mediates the relationship between X and Y, neither justifies the mediation effect nor contributes to theory development (Memon et al., 2018). Hence, the mediation effect hypothesis should be developed and articulated with theoretical and empirical justification (Rungtusanatham et al., 2014). Together with an explanatory purpose, several authors (Danks & Ray, 2018; Shmueli et al., 2016) have recently proposed that mediation analysis can also be tackled by adopting a prediction approach. However, this point requires further study, and future research should provide useful guidelines for conducting mediation analysis from a predictive point of view (Shmueli et al., 2019).

Some scholars may feel confused about how hypothesizing a mediation relationship (Memon et al., 2018). Rungtusanatham et al. (2014) recommended two approaches to formally theorize a mediation effect: *Transmittal Approach* and *Segmentation Approach*. The choice of the approach depends on whether there are established theories and previous empirical studies to support indirect effect or not. For instance, if we apply theories with a hierarchy of effects such as value-attitude-behavior, belief-desire-behavior, or stimulus-organism-response theories, some variables representing the middle layer (e.g., attitude, desire, or organism) can be mediator, and these theories can directly support indirect effect (i.e., sequential effect) and mediator. Therefore, in this case, we can apply the *Transmittal Approach* to develop a mediation hypothesis with the support of a theoretical framework (Rungtusanatham et al., 2014). As demonstrated in Table 1, when we have theoretical support for the indirect effect from literature, the *Transmittal Approach* is more appropriate. The hypothesis only needs to focus on the mediation effect (i.e., why and how the mediator transfers the effect of *IV* onto *DV*). Otherwise, *Segmentation Approach* should be applied. In such a case, the rationale and theoretical support for two effects (not necessarily hypotheses) should be provided: the effect of *IV* on the mediator and the effect of the mediator on *DV* (Ramayah et al., 2018; Rungtusanatham et al., 2014). Moreover, a similar logic for theoretical support of mediation hypothesis can be observed in MacKinnon, Fairchild, and Fritz (2007) and Baron and Kenny (1986), without mentioning about segmentation approach.

Table 1
Hypothesis development for the mediator.

	Segmentation Approach
H1	1- Theoretical support for the effect of *X* on *M* 2- Theoretical support for the effect of *M* on *Y* 3- Statement of the hypothesis for the mediation effect of *M* Mediation effect Example 1: *M* mediates the relationship between *X* and *Y* Example 2: *X* has an indirect effect on *Y* through *M*
	Transmittal Approach
H1	1- Theoretical support for indirect effect based on sequential theories Mediation effect Example 1: *M* mediates the relationship between *X* and *Y* Example 2: *X* has an indirect effect on *Y* through *M*

Source: Adapted from Ramayah et al. (2018).

2.2. Mediation analysis methods

Generally, there are two categories of analysis methods for mediation effect: implicit procedures and explicit procedures. The implicit procedures are traditional approaches that infer the mediation effect by a set of individual inferential tests of path relationships between the independent and dependent variables (Baron & Kenny, 1986; MacKinnon et al., 2002). According to Hayes (2018, p. 115), one of the common-sense principles of statistical inference is that claim should be justified by the 'quantification of the phenomenon most directly relevant'. However, the logic of implicit procedures is neither based on the estimation nor inference of indirect effect, which is most relevant to the mediation effect. On the contrary, the explicit procedures are more recent approaches that infer the mediation effect by an inferential test of indirect effect (MacKinnon et al., 2002; Preacher & Hayes, 2008; Rungtusanatham et al., 2014). According to this review's findings, combining the implicit procedures and explicit procedures is very common in tourism and hospitality literature. In most of these studies, the inferential test of indirect effect was conducted only after the *Causal Steps Approach*. Hayes (2018) believed that this approach is redundant and will weaken the power to identify the mediation effect.

2.2.1. The implicit procedures

The implicit procedures are typically represented by the traditional *Causal Steps Approach* (Baron & Kenny, 1986). This approach (also called the *Baron and Kenny Method*) became popular in the 1980s and is still used by many scholars nowadays. Historically, many mediation studies were built on this approach's logic (Hayes, 2018). According to Baron and Kenny (1986), the presence of the mediation effect must meet three conditions. The first condition is that the independent variable (*X*) must be an effect on the dependent variable (*Y*) to be mediated. Hence, the *Causal Steps Approach's* prerequisite is that *X* must significantly affect *Y* (total effect). Otherwise, the testing stops if this condition is not met. The second condition is that *X* must significantly affect the mediator (*M*). The third condition is that *M* must significantly affect *Y* controlling for *X*. If all the conditions are met, the direct effect of *X* (effect of *X* on *Y* after inclusion of *M*) is compared to the total effect of *X* on *Y*. If the total effect shrinks (the direct effect is smaller than the total effect) while the direct effect is not significant, then the effect of *X* on *Y* is wholly mediated by *M*. However, if the total effect shrinks and the direct effect is significant, then the effect of *X* on *Y* is partially mediated by *M*.

Although prevailing in the past, methodological researchers have increasingly criticized the *Causal Steps Approach* (Aguinis et al., 2017; Hayes, 2009; Rucker et al., 2011; Shrout & Bolger, 2002; Zhao et al., 2010), and its popularity is gradually fading (Hayes, 2018). Many researchers like Hayes (2018), Nitzl et al. (2016), and Rungtusanatham et al. (2014) simply advised scholars to abandon the *Causal Steps Approach* as awareness has been gradually raised in academia that this approach is not 'ideal both statistically and philosophically' (Hayes, 2018, p. 115).

Another commonly used implicit procedure for mediation analysis is the *Test of Joint Significance* (Leth-Steensen & Gallitto, 2015; Mallinckrodt et al., 2006). This approach requires that both the direct effects of *X* on *M* and *M* on *Y* be statistically significant without a test of total effect and direct effect of *X* on *Y*. However, both the *Causal Steps Approach* and the *Test of Joint Significance* neither quantify nor perform any inferential test for the indirect effect, which violates one of the above-mentioned common-sense principles of statistical inference (Hayes, 2018). Hayes (2009) argued that it is the indirect effect that matters in mediation analysis. Similarly, Zhao et al. (2010) also insisted that the inference of a mediation effect should be based on the estimation and inferential test of the indirect effect. The inferential test of indirect effect refers to the explicit procedures, which are implicit procedures' counterparts.

2.2.2. The explicit procedures

The alternative category, the explicit procedures focus on the significance test of the indirect effect, which mainly include the *Sobel Test* (Sobel, 1982), the *Distribution of Product Method* (MacKinnon, Fritz, et al., 2007; Williams & MacKinnon, 2008), *Bootstrapping Method* (Shrout & Bolger, 2002), and *Monte Carlo Simulation Method* (Preacher & Selig, 2012), among which the *Sobel Test* and *Bootstrapping Method* are the most widely used methods in the tourism and hospitality literature based on our review results.

Recommended by Baron and Kenny (1986), the *Sobel Test* (Sobel, 1982) (also called the *Normal Theory Approach*, or the *Product of Coefficients Approach*) is a formal approach for an inferential test of the indirect effect. This approach assumes the sampling distribution of indirect effect *ab* is normal (*a* and *b* denote the path from *X* to *M* and *M* to *Y*, respectively). With an estimate of the standard error, both the *p*-value and the confidence interval estimate of *ab* could be derived to support the indirect effect's significance. The *Sobel Test* is especially popular as it could be easily performed either manually or automatically through SEM software (e.g., AMOS, Mplus, LISREL) (MacKinnon, 2012; Preacher & Hayes, 2008) or PROCESS macro (Hayes, 2018; Preacher & Hayes, 2004) integrated into SPSS and SAS. Despite its popularity, the *Sobel Test* is not recommended and criticized by many contemporary scholars (Hayes, 2018; Hayes & Scharkow, 2013; MacKinnon et al., 2002) because of two shortcomings. First, this approach assumes that the sampling distribution of *ab* is normal, while this assumption usually does not hold for indirect effect, especially for a small sample size (Craig, 1936; MacKinnon et al., 2002). Hair et al. (2017) argued that the multiplication of *a* and *b* results in a non-normal distribution of their product even though they themselves are normally distributed. This argument is also justified by simulation research that the sampling distribution of indirect effect is irregular in most empirical studies (Bollen & Stine, 1990). The other disadvantage of the *Sobel Test* is that it is less powerful and accurate to detect indirect effect than other inferential approaches (Hayes & Scharkow, 2013; MacKinnon et al., 2004). Besides, Sosik et al. (2009) have provided additional arguments against using the *Sobel Test* together with the PLS-SEM technique. Although Hayes (2018) opposed the use of the *Sobel Test*, ironically, it is the user-friendly PROCESS macro developed by Preacher and Hayes (2004) that makes the *Sobel Test* so popular.

As a resampling approach, the *Bootstrapping Method* does not assume the normality of the sampling distribution and, therefore, is widely applied to many inferential tests (Wood, 2005). This method is beneficial for small sample sizes as the assumption of normality is usually violated (Good, 2013). The application of bootstrapping in mediation analysis is initiated by Bollen and Stine (1990) and popularized by MacKinnon et al. (2004) and Preacher and Hayes (2004, 2008). By this method, the empirical representation of the indirect effect's sampling distribution is obtained and then used to derive the *p*-value and the confidence interval estimate of *ab*. Hayes (2018) argued that the *Bootstrapping Method* is more powerful and accurate than the *Sobel Test* for the inference of the mediation effect as it better respects the nature of the sampling distribution of indirect effect. Three approaches, namely *Percentile, Bias-corrected*, and *Bias-corrected and Accelerated Method*, can be applied to construct the bootstrap confidence interval (Efron & Tibshirani, 1993; Preacher & Selig, 2012). Generally, *Bias-corrected* and *Bias-corrected and Accelerated Method* perform better than the *Percentile Method* (Hayes, 2018). However, evidence shows that the *Bias-corrected Method* can slightly elevate the possibility of detecting an indirect effect that does not exist (Type I error) in some circumstances (see Hayes & Scharkow, 2013).

Besides the *Sobel Test* and *Bootstrapping Method*, alternative approaches such as the *Monte Carlo Simulation Method* (MacKinnon et al., 2004; Preacher & Selig, 2012) and the *Distribution of Product Method* (MacKinnon, Fritz, et al., 2007) also performs exceptionally well for the inference of indirect effect. However, these two approaches are primarily interchangeable, as they typically result in the same inferences in most cases (Hayes & Scharkow, 2013). Some scholars may wonder which method is better for the inference of indirect effect. The answer depends on their tolerance for Type I (claiming a non-existent indirect effect) and Type II (failing to identify a real indirect effect) errors according to some studies evaluating and comparing the performance of these approaches (e.g., Hayes & Scharkow, 2013; Preacher & Selig, 2012; Williams & MacKinnon, 2008). The risk of Type I error is lowest for the *Sobel Test,* but this approach is so conservative that the real indirect effect is more likely to be missed (Hayes, 2018). To ensure an acceptable level of Type I error while maintaining enough statistical power, the *Percentile Bootstrapping Method*, the *Monte Carlo Simulation Method*, or the *Distribution of Product Method* are generally suggested (Cheung & Lau, 2007; Hayes & Scharkow, 2013). In fact, because of its wide availability in much statistical software such as PROCESS macro, AMOS, and SmartPLS (Memon et al., 2018; Zhao et al., 2010), the *Percentile Bootstrapping Method* is more common in mediation analysis, which is consistent with our review findings of tourism and hospitality literature. In addition, Aguirre-Urreta and Rönkkö (2018) have also recommended the *Percentile* approach for statistical inference in PLS-PM.

2.3. The interpretation of mediation effect

2.3.1. Complete and partial mediation

Once having detected mediation effect through an inferential test of the indirect effect, some scholars tend to further classify the effect into complete (full) or partial mediation according to the significance of direct effect (*c*') of *X* on *Y*. If *c*' is not significant, then it can be concluded that *M* completely mediates the effect of *X* on *Y*; otherwise, *M* only partially mediates the effect (Baron & Kenny, 1986) (Fig. 2). Some scholars (*e.g.*, Hayes & Rockwood, 2017; Rucker et al., 2011) believed that it is fruitless to distinguish between full and partial mediation, and this concept should be abandoned. Because the sample size strongly influences the significance of the direct effect (*c*'), Nitzl et al. (2016) recommended that researchers have to exercise some caution when talking about full mediation in the case of small samples. In this vein, Rucker et al. (2011, p. 364) highlighted, 'the smaller the sample, the more likely mediation (when present) is to be labelled full as opposed to partial because *c*′ is more easily rendered nonsignificant'. On the other hand, Rucker et al. (2011) illustrated in their study that claiming one mediator completely mediates the effect does not preclude other researchers from identifying different mediators with the same complete mediating effect. Rucker et al. (2011) further argued that one could never claim full mediation as in social science, it is practically

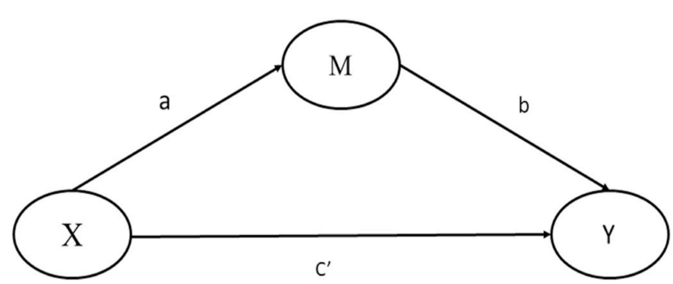

Fig. 2. Mediation model.

impossible to identify and measure all the potential mediators. Hence, Hayes (2018) concluded that full mediation is an empty concept, and there is no point in claiming it. Notwithstanding, an endogeneity analysis of the model could help identify omitted variables (Hult et al., 2018), the primary source of endogeneity (Bascle, 2008). An omitted construct might affect both the *DV* and one or more *IVs* in the model. As a result, path coefficients estimates could be biased and inconsistent, making them causally uninterpretable and potentially causing Type I and Type II errors (Papies et al., 2017). Therefore, the identification of an endogeneity problem would imply the impossibility of sustaining the existence of a complete mediation.

Furthermore, Hayes (2018) also advocated against a claim of partial mediation. He argued that partial mediation implies a misspecified model where the mediator has not explained at least part of *X*'s effect. On the other hand, Zhao et al. (2010., p. 199) believed that the sign of direct effect *c*' is a 'silver lining' inside the partial mediation model, which is heuristic for the researchers to explore new mediators and thus contributes to theory development.

2.3.2. Report of mediation effect

Now that only the significance of indirect effect *ab* matters in the mediation analysis (Hayes, 2018; Preacher & Hayes, 2008) and the claim of complete or partial mediation (depending on the significance of direct effect *c'*) is of little value according to some scholars (e.g., Hayes & Rockwood, 2017; Memon et al., 2018; Rungtusanatham et al., 2014), is it still necessary to consider the signs and significance of *a*, *b*, and *c'*?

Hayes (2018) and Zhao et al. (2010) argued that interpreting the indirect effect without attention to the signs of *a* and *b* may draw erroneous conclusions. They explained that one might predict a positive indirect effect because both *a* and *b* are hypothesized to be positive based on the theory. However, a positive indirect effect can still be yield even if both *a* and *b* are negative, which is opposite to the mechanism that the theory explains.

Moreover, Zhao et al. (2010) and Rungtusanatham et al. (2014) also advised scholars to report both the signs and significance of *c'* to interpret the mediation results better. Zhao et al. (2010)'s main contribution is that they proposed a typology of mediation and further categorized Baron and Kenny (1986)'s partial mediation into complementary (*abc'*, the product of the indirect effect *ab* and direct effect *c'*, is positive) and competitive mediation (*abc'* is negative). Because *X*'s total effect on *Y* is the sum of the indirect effect *ab* and direct effect *c'*, in some cases, *ab* and *c'* will offset each other if their signs are opposite. Therefore, the sign and significance of *ab* and that of *c'* are of great importance to the interpretation of the mediation results. More importantly, to report the sign and significance of the direct effect is a heuristic for exploring omitted mediators and theory building (Zhao et al., 2010).

On the other hand, given the relevance of statistical significance decreases in social science because of the increasing availability of large data sets, Mohajeri et al. (2020) have recently urged paying attention to the notions of relevance and practical significance. For this reason, it could be pertinent to consider the variance accountant for (VAF) value in the case of complementary mediation (Henseler, 2021). The VAF is defined as the ratio of the indirect-to-total effect. Thus, VAF would help determine the portion of the total effect attributable to the indirect effect and be reported as a percentage value. Henseler (2021) shows an interesting practical example of using VAF. The total effect of an independent on a dependent variable is decomposed into direct and indirect effects using a pie chart together with percentage values. Through this way, the relevance of the indirect effect can be quickly grasped.

3. Methods

This systematic review was conducted to evaluate the assessment methods of mediation models in tourism and hospitality academia. The review was performed following the reporting checklist of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA; Liberati et al., 2009). Because of the research mentioned above objective, a protocol was developed to reduce the biases and improve the reliability of this review (Pahlevan-Sharif et al., 2019); it specified the review procedures such as information sources (e.g., databases), retrieve terms, eligibility criteria, and study selection processes.

Social Sciences Citation Index (SSCI) database was selected to retrieve the relevant literature. The justification for only selecting SSCI journals lay on the research objective that this review aimed to evaluate the trend of mediator assessment methods and thus suggest new guidelines for future studies. Therefore, journal quality, rather than comprehensiveness, was prioritized in this review. Besides, in the discipline of tourism and hospitality, 16 SSCI journals (21 in total) almost overlap with the A* and A journals (20 in total) in the Australian Business Deans Council (ABDC) journal quality list (based on the rank of 2019), which implies the recognized quality of SSCI journals in this discipline. To capture studies that had assessed the mediating effect, "mediat*" was applied as the retrieve term. The asterisk (*) was used as the wildcard because it represented any combination of characters. In this way, "mediat*" could match such words as "mediator(s)", "mediation", "mediating", and "mediated". The term "indirect*" was initially included to identify expressions like "indirect relationship", and "… is affected (influenced) indirectly by …", but then omitted as it resulted in excessive irrelevant articles.

As of February 2021, based on the advanced search function of Web of Science, TI (topic) tag was applied to retrieve articles containing the term "mediat*" in their title, abstract, author keywords, and keywords plus. Language and document type were confined to English and article, respectively, while no timeframe was imposed. The results were further refined by Hospitality Leisure Sport Tourism category according to Web of Science categories. Also, sport journals in this category were excluded. This resulted in 1545 records in total, 75.66% of which were from 2016 to 2020. Again, the purpose of this review was to evaluate the latest trend of methodological choice and present new guidelines for future studies, while comprehensiveness was not the priority. Therefore, 1169 studies from 2016 to 2020 were selected, which was adequate to fulfil the aforementioned review objective.

Scholars typically first screen the titles and abstracts of the retrieved studies and exclude those irrelevant records (Pickering & Byrne, 2014), primarily when the systematic review is concerned with specific topics or themes. However, as this review focused on the mediator assessment's methodological choice, merely screening the titles and abstracts was inadequate to refine the eligible studies further. Therefore, instead of titles and abstracts, the retrieved articles' full texts were directly reviewed for eligibility to be included. All the articles were imported into NVivo 12 for further screening. In aid of the text search function of NVivo, relevant contents relating to mediator were located in each article by using the search term "mediat*". Then two independent authors checked the predefined eligibility criteria of each article. Disagreements concerning article exclusion were discussed and settled by consensus to avoid improper exclusion of eligible studies (Pahlevan-Sharif et al., 2019). In this stage, studies that did not meet the eligibility criteria were discarded. The screening finally yielded 536 studies, of which full texts were thoroughly reviewed to extract the relevant data.

The title, author names, year of publication, journal name, and abstract of these 536 records were tabulated in a Microsoft Excel spreadsheet for data management. Before extracting data from the identified article, a pilot test comprised of 50 randomly selected articles was conducted by two reviewers independently to refine the categories for data extraction. Disputes regarding the extraction categories were settled by consensus between the two reviewers. Once the categories were confirmed, the eligible articles were then scrutinized to extract the relevant information by one reviewer. Although reviewed only by one researcher, the review's reliability would not be significantly influenced as the data required to be extracted in this review was objective and unique, where subjective coding for qualitative information was unnecessary. NVivo database was created for data mining based on the full text of all the eligible articles. With the aid of NVivo's text search function, relevant information could be located and extracted more efficiently. The text search strings were as follows: mediat* OR hypothe* OR theor* OR model OR Hayes OR "Baron & Kenny" OR "Baron and Kenny" OR Sobel OR bootstrap* OR "total effect" OR indirect* OR direct* OR regression OR "process macro" OR SEM OR "structural equation" OR AMOS OR SPSS OR MPLUS OR LISREL OR PLS OR "partial least" OR EQS.

4. Review findings

536 articles met the eligibility criteria, among which 308 studies (57.46%) were in the hospitality discipline and 228 (42.54%) in the tourism discipline. As shown in Table 2, studies that had assessed mediation effect in the hospitality and tourism field were mainly published in three journals, namely *International Journal of Contemporary Hospitality Management* (n = 106), *International Journal of Hospitality Management* (n = 84), and *Journal of Hospitality and Tourism Management* (n = 42), particularly before 2017, as about half of the remaining journals only published one or two such articles while others did not publish any in 2016. Since 2017, the assessment of the mediation effect had attracted the attention of many hospitality and tourism scholars. Published studies had increased dramatically, especially in the following journals, namely *Current Issues in Tourism*, *Tourism Management Perspectives*, *Journal of Hospitality Marketing & Management*, *Journal of Travel & Tourism Marketing*, and *Asia Pacific Journal of Tourism Research*, as more than 97.04% of papers that had assessed mediation effect in the above-mentioned journals were published in the recent four years. The

Table 2
A summary of the Journal names and year of publication of the eligible articles.

Journal Names	2016	2017	2018	2019	2020	Total
International Journal of Contemporary Hospitality Management	16	20	25	26	19	106
International Journal of Hospitality Management	10	10	6	18	40	84
Journal of Hospitality and Tourism Management		5	4	9	24	42
Tourism Management	10	6	6	6	3	31
Current Issues in Tourism	1	2	2	4	21	30
Tourism Management Perspectives	1	2	4	10	11	28
Journal of Hospitality Marketing & Management	1	2	3	11	10	27
Journal of Travel & Tourism Marketing	1	5	11	5	4	26
Asia Pacific Journal of Tourism Research		6	5	4	9	24
Journal of Hospitality & Tourism Research	1	1	3	6	11	22
Journal of Sustainable Tourism	1		4	2	15	22
Journal of Destination Marketing & Management	2	1	1	2	11	17
Journal of Travel Research		3	3	2	9	17
Tourism Review	1	3	2	4	4	14
Cornell Hospitality Quarterly	2	1	4	2	3	12
International Journal of Tourism Research		2	1	3	5	11
Journal of Vacation Marketing			1	3	4	8
Journal of Hospitality and Tourism Technology			4		3	7
Tourism Economics			1		1	2
Annals of Tourism Research			1	1		2
Tourism Geographies		1			1	2
Journal of Tourism and Cultural Change					1	1
Scandinavian Journal of Hospitality and Tourism				1		1
Total	**47**	**70**	**91**	**119**	**209**	**536**

characteristics of the eligible studies (including types of mediation, number of mediators, statistical methods, and applied software), and the details of mediation assessment (including features of mediation hypothesis, mediation assessment methods, the way to test significance of indirect effect, reference to full or partial mediation, and reference to complementary or competitive mediation) were reviewed and quantified in the following sections.

4.1. *The characteristics of the eligible studies*

The types of mediation models and the number of mediators are summarized in Table 3. The most frequently used model by scholars was the simple mediation model, which accounted for nearly half (n = 250, 46.64%) of the studies, followed by the parallel multiple mediation (mediators operate in parallel, without affecting each other) model (n = 141, 26.31%) and serial multiple mediation (mediators are linked together in a causal chain) model (n = 104, 19.4%). Scholars rarely used the multiple mediation model that incorporated both parallel and serial mediators, and only 41 (7.65%) studies applied this model. Regarding the number of mediators in the model, scholars tended to include less than four mediators, as 93.62% of parallel multiple mediation models and 95.19% of serial multiple mediation models included only two or three mediators.

Mediation studies using regression and structural equation modelling (SEM) approaches were the focus of this review. As shown in Table 4, about two-thirds of the reviewed studies (n = 341, 63.62%) applied covariance-based structural equation modelling (CB-SEM) to assess mediation effect, followed by partial least squares structural equation modelling (PLS-SEM) (n = 115, 21.46%) and linear regression (n = 80, 14.93%). AMOS (71.48%) was the prevailing software among scholars who applied CB-SEM, while Mplus (14.08%) and LISREL (10.92%) are less popular. The PLS-SEM studies were dominated by SmartPLS, with a percentage of 95.15%. The remaining studies based on linear regression were mainly performed using SPSS or PROCESS (Hayes, 2018), which could be integrated as a macro in SPSS and SAS. Overall, the most common software used by hospitality and tourism scholars for mediation assessment were AMOS, SmartPLS, and PROCESS macro.

4.2. *The details of mediation assessment*

4.2.1. *Features of mediation hypotheses*

As shown in Fig. 3, 81.72% (n = 438) of the studies had hypothesized the mediation effect concerning the mediation hypothesis. However, the remaining studies (n = 98, 18.28%) did not include any mediation hypothesis but assessed the mediation effect in the results section. To justify the mediation effect, only 38.58% (n = 169) of the studies substantiate the exogenous variables' indirect effects on endogenous variables through the mediator. In contrast, most of the studies (n = 269, 61.42%) merely demonstrated the mediation effect by discussing two direct effects: the exogenous variable's effect on the mediator and the mediator's effect on the endogenous variable. The results show that most scholars inappropriately used the *Segmentation Approach* because the indirect effect was not justified theoretically or empirically. It is found that most of the studies (n = 281, 64.16%) justified the relationships only through empirical studies, while 35.84% (n = 157) also drew on theory besides empirical evidence.

4.2.2. *Mediation analysis procedures*

Mediation analysis methods can be generally classified into two categories: implicit procedures and explicit procedures. As shown in Table 5, the explicit procedures dominate in mediation assessment as they account for 74.63% (n = 400) of the reviewed studies. Studies that applied the implicit procedures and combined both procedures (testing the indirect effect after conducting causal steps approach) are 11.57% (n = 62) and 12.69% (n = 68), respectively. Furthermore, as illustrated

Table 3
Characteristics of mediation models.

Mediation Models	Number of Mediators						
	1	2	3	4	5	6	Total
simple mediation	250 (100%)						250 (46.64%)
parallel multiple mediation		107 (75.89%)	25 (17.73%)	4 (2.84%)	4 (2.84%)	1 (0.71%)	141 (26.31%)
serial multiple mediation		79 (75.96%)	20 (19.23%)	4 (3.85%)	1 (0.96%)		104 (19.4%)
multiple mediation (parallel & serial)			28 (68.29%)	8 (19.51%)	4 (9.76%)	1 (2.44%)	41 (7.65%)
Total	**250**	**186**	**73**	**16**	**9**	**2**	**536**

Table 4
Statistical methods and software.

Statistical methods	Software	Number of studies	Percentage
CB-SEM	AMOS	203	71.48%
	Mplus	40	14.08%
	LISREL	31	10.92%
	EQS	8	2.82%
	Stata	2	0.7%
	not mention	57	
		341	**63.62%**
PLS-SEM	SmartPLS	98	95.15%
	XLSTAT	3	2.91%
	Warp PLS	2	1.94%
	not mention	12	
		115	**21.46%**
Regression	PROCESS macro	57	77.03%
	SPSS	15	20.27%
	R	2	2.7%
	not mention	6	
		80	**14.93%**
Total		**536**	**100.00%**

in Fig. 4, the trend of different methods applied between 2016 and 2020 also shows that the number of studies using the explicit procedures has increased steadily, while the curves of other methods have remained even. Among the studies that have applied the explicit procedures, about 59% (n = 236) of them tested and reported both the indirect and direct effect, while the remaining 41% (n = 164) only considered the indirect effect. Among these 164 studies, 42.68% (n = 70) did not include the path of direct effect in the conceptual framework. It should be noted that in order to test a mediation effect between an independent variable (*X*) and a dependent variable (*Y*) via a mediator (*M*), the direct effect (*c*') between *X* and *Y* must be present in the model. If this is not carried out, as is the case in the 70 cases above, it would mean an error in the estimation because the *X*'s effect on *Y* is not controlled (Nitzl et al., 2016), so the indirect effect estimate could be inflated in these studies. Regarding those studies that have applied the implicit procedures, most of them (n = 53, 85.48%) used the causal steps approach, while only a few of them (n = 9, 14.52%) used the test of joint significance approach.

4.2.3. Inference of the indirect effect

Regarding the inference of the indirect effect, Table 6 shows that most scholars in tourism and hospitality have applied the *Bootstrapping Method* (n = 346, 73.93%), followed by the *Sobel Test* (n = 108, 23.08%), the *Monte Carlo Simulation Method* (n = 10, 2.14%), and others (n = 4, 0.85%). Among those applying the *Bootstrapping Method*, 43.06% (n = 149) and 38.73% (n = 134) have reported the bootstrap confidence interval and bootstrap *p*-value, respectively, and the remaining 18.21% (n = 63) have reported both the bootstrap confidence interval and *p*-value. To estimate the bootstrap confidence interval, as shown in Fig. 5, more than half of the studies (n = 136, 56%) used the *Percentile Method*, and 37% (n = 90) used the *Bias-corrected Method*. In comparison, only 3% (n = 6) used the *Bias-corrected and Accelerated Method*. Among those applying the *Sobel Test* (n = 108), 34.26% (n = 37) have also used different bootstrapping methods to cross-validate the results. Only a small proportion of the studies have used the *Monte Carlo Simulation Method* (n = 10) and other methods (n = 4). As illustrated in Fig. 6, before 2017, there was no significant difference in applying these methods. However, after 2017, the bootstrapping method's application has dramatically increased, while the *Sobel Test*'s popularity has decreased, given the increased number of mediation studies in recent years.

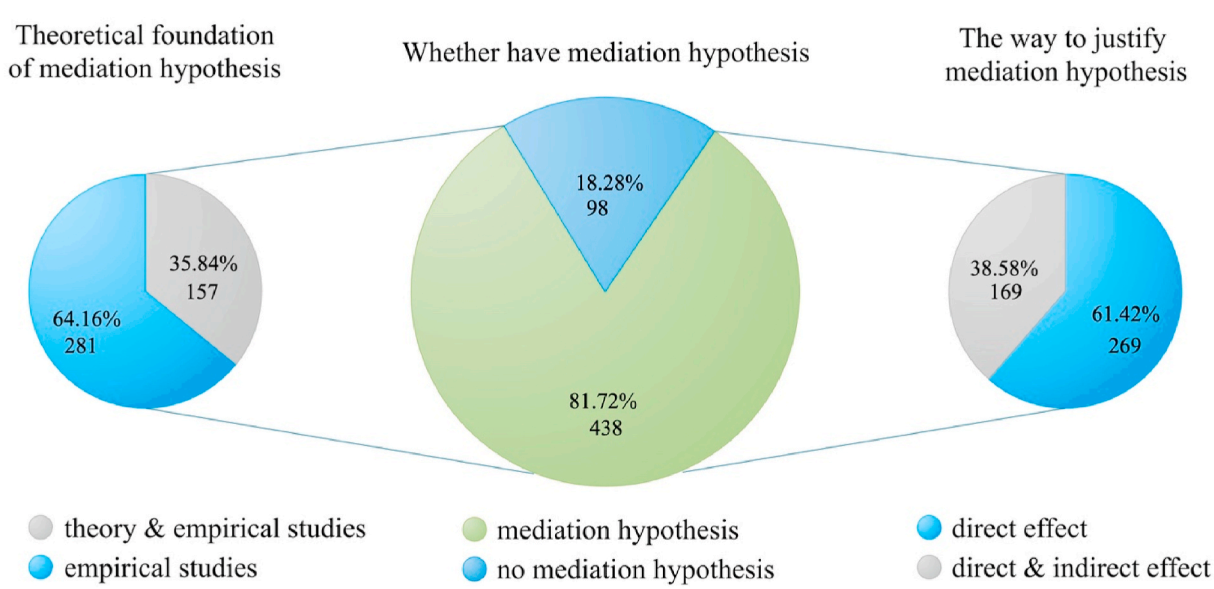

Fig. 3. Features of mediation hypothesis.

Table 5
Overview of mediation analysis methods.

Mediation analysis methods		2016	2017	2018	2019	2020	Total
The explicit procedures	indirect and direct	25	26	41	62	82	236 (59%)
	only indirect effect	3	18	22	32	89	164 (41%)
	—no path of direct effect in framework	1	12	10	16	31	70
		28	**44**	**63**	**94**	**171**	**400 (74.63%)**
The implicit procedures	causal steps approach	9	10	13	8	13	53 (85.48%)
	test of joint significance		4	1	2	2	9 (14.52%)
		9	**14**	**14**	**10**	**6**	**62 (11.57%)**
Combination of implicit and explicit procedures	test indirect effect after the causal steps approach	**10**	**11**	**13**	**14**	**20**	**68 (12.69%)**
Others	comparison of model fit		**1**	**1**	**2**	**2**	**6 (1.12%)**
Total		**47**	**70**	**91**	**120**	**208**	**536**

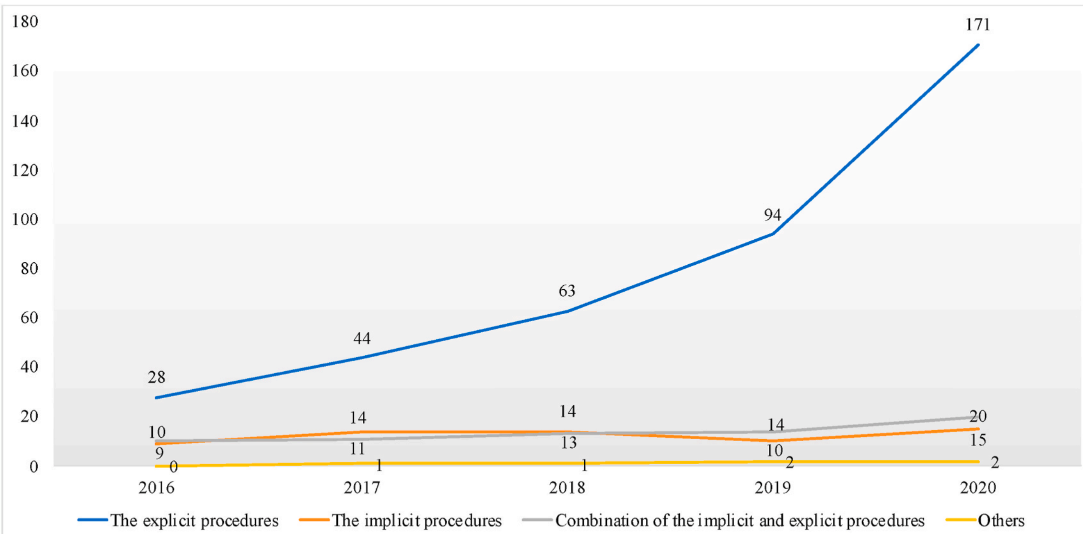

Fig. 4. The trend of mediation analysis methods.

Table 6
Overview of inferential methods of the indirect effect.

Inferential test of indirect effects	2016	2017	2018	2019	2020	Total
Total of bootstrapping methods	23	33	52	76	162	346 (73.93%)
Bootstrap confidence interval	8	11	20	36	74	149 (43.06%)
Bootstrap *p*-value	12	15	26	28	53	134 (38.73%)
Bootstrap *p*-value & confidence interval	3	7	6	12	35	63 (18.21%)
Total of Sobel test	**15**	**19**	**22**	**29**	**23**	**108 (23.08%)**
Sobel test	10	14	15	18	14	71 (65.74%)
Sobel & Bootstrap confidence interval	4	4	5	7	9	29 (26.85%)
Sobel & Bootstrap *p*-value	1	1	2	4	0	8 (7.41%)
Monte Carlo confidence intervals	**0**	**2**	**0**	**3**	**5**	**10 (2.14%)**
Others	**0**	**1**	**2**	**0**	**1**	**4 (0.85%)**
Total	**38**	**55**	**76**	**108**	**191**	**468**

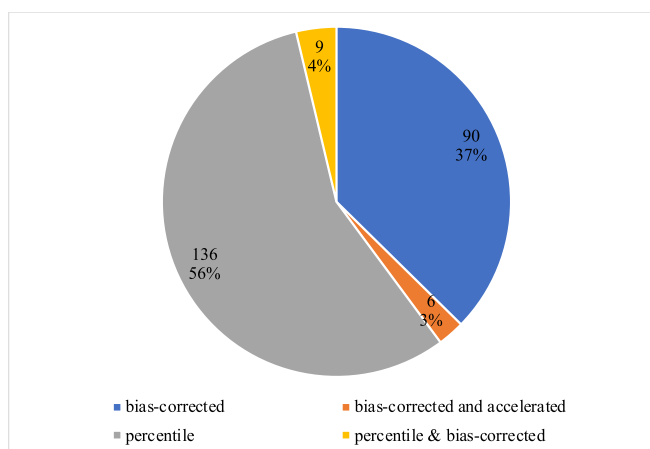

Fig. 5. Types of bootstrap confidence intervals.

4.2.4. Interpretation of the mediation effect

More than half (n = 280, 52.24%) of the studies have claimed the mediation effect as complete (full) or partial mediation. Even though some scholars (e.g., Hayes & Rockwood, 2017; Rucker et al., 2011) argued that the concept of complete and partial mediation should be abandoned, this concept is still prevalent in the literature of tourism and hospitality. Furthermore, 16 studies have reported variance accounted for (VAF) value of mediator to evaluate partial mediation's strength. Surprisingly, only 22 studies have used Zhao et al. (2010)'s typology to distinguish between competitive and complementary mediation.

5. Discussion

This paper aims to provide guidelines based on the lessons from the review of published articles in top tier hospitality and tourism journals,

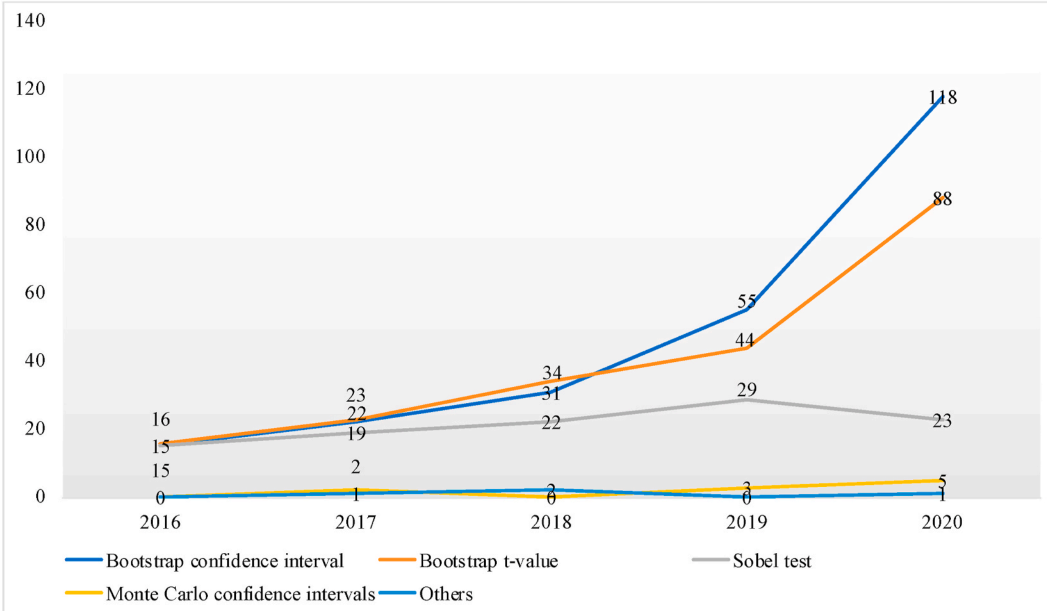

Fig. 6. The trend of the inferential methods of the indirect effect.

which have analyzed mediation and recent literature regarding mediation analysis practices. According to the review results, 18.28% of studies published in top tier tourism and hospitality journals have not hypothesized the mediation effect, and only assessed the mediator in the analysis stage. The majority of the articles (81.72%) had hypothesized the mediation effect. However, only 35.84% discussed theoretical support for mediation effect using either *Transmittal* or *Segmentation* approaches to develop mediation hypotheses. For quantitative research, supporting hypotheses by theory is critical, and mediation hypotheses are not exceptional (Fisher & Aguinis, 2017; Hair et al., 2017; Memon et al., 2018). Therefore, the first and most crucial step for mediation analysis is to develop mediation hypotheses based adequately on theoretical support and literature (MacKinnon et al., 2012; Rungtusanatham et al., 2014). As mentioned earlier, theoretical support for mediation hypotheses can be based on the *Transmittal Approach*, if there is a theory that can support indirect effect such as sequential theories (e.g., the value-attitude-behaviour or stimulus-organism-response theories), and there are previous studies, which hypothesized and tested indirect or mediation effect (Ramayah et al., 2018; Rungtusanatham et al., 2014). However, if there is no theoretical support for indirect effect and mediation in the literature, the *Segmentation Approach* can be applied (Ramayah et al., 2018; Rungtusanatham et al., 2014). In *the Segmentation Approach*, we need to provide support from theory and literature for the effect of *IV* to Mediator and the effect of Mediator on *DV*. This should be noted that in the *Segmentation Approach*, it is unnecessary to have those hypotheses in the study. Instead, theoretical support for those relationships is required.

Statistical assessment of mediation effect is the next step in the process of mediation analysis, which needs full consideration of researchers. Application of the incorrect statistical method to assess mediation leads to biased results. According to the review results, the majority of the studies (74.63%) have applied an explicit approach and statistically tested the significance of the indirect effect. However, 24.26% of studies have still applied the implicit approach, which several studies have highlighted the low power and biasness of results of this category of statistical tests to assess mediators (Hayes & Scharkow, 2013; MacKinnon et al., 2002). It is worth noting that the number of studies using the explicit approach has increased steadily in recent years, particularly, 82.2% of the studies applied the explicit approach in 2020. Unfortunately, the flawed implicit approach was still be applied by some researchers. Future researchers should abandon this outdated approach.

On the other hand, we should warn of a dangerous practice that we have detected among those papers that use explicit procedures but only analyze the indirect effect. We refer to studies that focus on analyzing a mediation effect without including in the model the direct effect between the IV and the DV. This approach implies incurring a notable bias in the estimation of the indirect effect since this could be inflated due to not having controlled the effect of *X* on *Y*. Consequently, this could lead to an increase in the probability of the existence of Type I error. Out of 468 studies, which applied explicit approach (e.g. the product of coefficients approach), however, 108 articles employed the *Sobel Test* to check the significance of the indirect effect, while the *Sobel Test* has been identified as one of the approaches to test mediator with low power (Hayes & Scharkow, 2013). The review results showed that a majority of articles had applied bootstrapping resampling using different methods and criteria such as *p*-value, percentile confidence interval, bias-corrected, and acceleration confidence interval. In recent years, 2020 witnessed a significant increase in the studies applying the bootstrapping resampling method to test the mediation effect. Unfortunately, some researchers still apply outdated and low power *Sobel Test*, which future researchers should also abandon. Based on recent developments and recommendations for mediation analysis, a researcher in hospitality and tourism field should consider the application of explicit approach (e.g., the product of coefficients) using bootstrapping resampling method, and report percentile, bias-corrected, and acceleration confidence interval to get most robust results for assessment of mediator and indirect effect.

For interpretation of results of the assessment of mediation effect, more than 50% of reviewed articles have reported full and partial mediation, which recent literature (e.g., Hayes & Rockwood, 2017; Rucker et al., 2011) believed that the concept of full and partial mediation is not meaningful. Full mediation cannot be claimed only because of the insignificance of direct effect. There is always the possibility of other mediators' existence for transferring the effect of *IV* to *DV*. Rucker et al. (2011) argued that one could never claim complete mediation as in social science, it is practically impossible to identify all potential mediators. Moreover, Hayes (2018) argued that the concept of partial mediation implies a misspecified model where at least part of the effect of *IV* is transmitted to *DV* through a mediator. However, recent literature still recommends the importance of discussion and comparison of direct and indirect effects for theory development and advancement (Nitzl et al., 2016; Zhao et al., 2010). Hayes (2018) and Zhao et al. (2010) suggested comparing and interpreting size and sign of the direct and indirect effect. Zhao et al. (2010) proposed a typology of mediation, namely complementary (the product of the indirect effect *ab* and direct effect after including mediator *c'*, have same signs) and competitive mediation (the product of the indirect effect *ab* and direct effect after including mediator *c'*, have opposite signs). Out of 536 reviewed articles, only 22 articles reported and discussed the sign of direct and indirect effects using the Zhao et al. (2010)'s approach. Therefore, hospitality and tourism researchers need to interpret the sign of indirect and direct effects to understand better the mechanism of the effect of *IV* to *DV*. This interpretation can contribute to theory advancement by highlighting the type of mediator and the possibility of other available mediators, mainly when the moderator is competitive compared to a complementary mediator.

6. Conclusion

This study aims to provide guidelines for researchers in the tourism and hospitality field and other social science disciplines for mediation analysis including mediation hypothesis development, statistical assessment of mediation effect, and interpretation of mediation assessment results. Table 7 shows these three main steps based on lessons from the review of 536 articles published in tourism and hospitality journals and recent literature on mediation analysis. **Step 1** refers to hypothesis development for the mediation effect. In this step, we need to develop a mediation hypothesis by supporting theory and literature using the *Transmittal Approach* or the *Segmentation Approach*. For instance, suppose a unified theory, a sequential theory such as the value-attitude-behaviour theory, is available to support indirect effect, and previous literature tested and confirmed the mediation effect. In that case, the indirect effect or mediation can be hypothesized. However, if there is no direct support from theory and previous studies for indirect effect, so using the *Segmentation Approach,* we need to provide theoretical support for three effects: 1) effect IV to M, and 2) effect of M to DV. **Step 2** refers to the statistical assessment of the mediator. Recent literature recommends applying the product of the coefficients approach using bootstrapping resampling method and percentile, bias-corrected, and acceleration confidence interval to test the significance of the indirect effect. Finally, in **Step 3**, to interpret the results of mediation analysis, using Zhao et al. (2010)'s approach, researchers need to interpret the sign of indirect and direct effects to understand better the mechanism of the effect of *IV* to *DV*. If the sign of indirect effect (*ab*) and direct effect after including mediator (*c'*) are the same the mediation effect is complimentary; however, if the sign of indirect effect (*ab*) and direct effect (*c'*) are opposite, the mediation effect is competitive. In the case of complementary mediation, it may be appropriate to examine the variance accountant for (VAF) value (Henseler, 2021), that is, the indirect-to-total effect ratio. As a result, VAF would help determine the percentage of the total effect attributed to the indirect effect, which would then be reported as a percentage number. Then, following Henseler (2021), using a pie chart and these percentage values, the overall influence of an independent variable on a dependent variable is divided into direct and indirect effects. Therefore, the importance of the indirect effect can be immediately comprehended this way.

Table 7
Guidelines for mediation analysis.

Steps for mediation analysis	Description
Step 1: Hypothesis development H1: "*M* mediates the relationship between *X* and *Y*" OR "*X* has an indirect effect on *Y* through *M*"	1 Using the *Transmittal Approach*, if, a suitable theory, for instance, a sequential theory such as the value-attitude-behavior theory is available to support indirect effect, and previous literature tested and confirmed the mediation effect. 2 Using the *Segmentation Approach*, if there is no direct support from theory and previous studies for the indirect effect. In this case, we need to provide theoretical support for two effects: 1) effect *IV* to *M*, and 2) effect of *M* to *DV*.
Step 2: Statistical assessment	Refer to Fritz and Mackinnon (2007) for the adequate sample size for mediation analysis. Apply the product of coefficients approach using bootstrapping resampling method, and percentile, bias-corrected, and acceleration confidence interval to test the significance of the indirect effect.
Step 3: Interpretation	Using Zhao et al. (2010)'s approach, interpret the sign of indirect and direct effects. 1 If the sign of indirect effect (*ab*) and direct effect after including mediator (*c*) are the same, the mediation effect is complementary. • In this case, consider examining the VAF value, which will describe the portion of the total effect attributable to the indirect effect (Henseler, 2021). 2 If the sign of indirect effect (*ab*) and direct effect (*c*) are opposite, the mediation effect is competitive.

6.1. Theoretical and methodological implications

This paper by providing a robust guideline for mediation analysis based on reviewing the published articles in top tier journals and recent development on mediation analysis literature, has significant theoretical and methodological contribution in the tourism and hospitality field and social science disciplines in general. As the literature review revealed, mediators in the research framework are significantly increasing in recent years. However, several critical issues in different mediation analysis steps, including hypothesis development, statistical assessment, and interpretation of results, cause limited and unreliable understanding of mediation results and questionable findings for theory building and development. The aim for the inclusion of mediator in a framework is to understand the mechanism of the effect of *IV* to *DV* and so understanding and advancement of theory. If researchers follow the incorrect process from hypothesis developments to the interpretation of results, theory development and advancement will fail.

In practice, many researchers who apply the mediation analysis tend to follow previous empirical literature or ancient methodological literature. In contrast, few researchers are willing to delve into the up-to-date recognized methodological literature. Similarly, some reviewers, as researchers themselves, often fail to recognize the methodological flaws of manuscripts. Undeniably, under the influence of previous empirical literature, the robust approaches will be gradually applied, corroborated by our review results. Although researchers have criticized the flaws of the implicit approach and the Sobel Test for more than one decade (e.g., Hayes, 2009; Rucker et al., 2011; Zhao et al., 2010), not until recent years did tourism and hospitality researchers begin to adopt the robust approaches to test mediation effect widely.

Given that some researchers are still applying some outdated approaches, we believe that the speed of disseminating these robust approaches is equally important. Otherwise, biased findings and unreliable interpretations caused by the continued application of outdated approaches and approaches that are not correctly applied would diminish the quality of publications. Therefore, this paper is helpful for tourism and hospitality researchers to realize the weaknesses of those outdated approaches and abandon them.

Although more and more researchers have begun to adopt more robust approaches regarding the statistical test of the mediation effect in recent years, they still need to take a more robust way regarding mediation hypotheses development and results interpretation. Nowadays, there is a tendency in social sciences. Some researchers tend to include mediator(s) or moderator(s) without solid theoretical foundations to complicate the model and assume that it can make significant contributions. However, this tendency is problematic because failing to provide theoretical support for the mediation hypothesis would make the following mediation analysis meaningless. Our review also shows that many studies in tourism and hospitality lacked theoretical support for hypothesis development of mediation effect, which is the first and most crucial step for mediation analysis. If researchers decide to investigate the mediation effect, they should seriously think about the theoretical rationale and practical significance. In addition, when interpreting the results of mediation analysis, more than half of the studies have still reported full or partial mediation. In contrast, only 22 studies have used Zhao et al. (2010)'s approach to interpret the sign of indirect and direct effects, which can significantly contribute to theory advancement. Therefore, the current practice regarding mediation hypotheses development and results interpretation needs to be further improved in tourism and hospitality.

More importantly, researchers should know what approaches they should use and how to use them by simply following previous literature and understand why the applied approaches are more advantageous. Therefore, providing a robust guideline for mediation analysis by synthesizing those recognized methodological approaches scattered in various literature has a significant theoretical contribution to the field. Besides, this paper, by reviewing many papers published in top tier tourism and hospitality journals, highlighted the common methodological mistakes for mediation analysis. By highlighting those common methodological mistakes and misconceptions, researchers can apply a more robust methodology for assessing and interpreting the mediation effect by avoiding those mistakes and misconceptions.

6.2. Limitations and future research

This research only reviewed the articles published in SSCI tourism and hospitality journals. Therefore, the results of the review are limited to tourism and hospitality journals. Future research can be conducted in other social science disciplines to have a more comprehensive picture of the application of mediator, hypothesis development, statistical assessment, and interpretation.

Declaration of competing interest

None.

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